

Question

An ROC curve plots:

- A. Precision against Recall
- B. True Positive Rate against False Positive Rate
- C. Accuracy against Threshold
- D. Recall against Precision

Answer: B

True Positive Rate against False Positive Rate

Explanation

An ROC curve (Receiver Operating Characteristic) plots the True Positive Rate (TPR) against the False Positive Rate (FPR) across all possible classification thresholds:

$$\text{TPR} = \frac{TP}{TP + FN} = \text{Sensitivity} / \text{Recall}$$

$$\text{FPR} = \frac{FP}{FP + TN} = 1 - \text{Specificity}$$

How to read it:

The top-left corner (TPR = 1, FPR = 0) is perfect classification.

The diagonal line (TPR = FPR) represents random guessing.

A curve closer to the top-left corner is a better model.

Why the others are wrong:

A: Precision vs. Recall is the *Precision-Recall curve*, a different plot (often better for imbalanced data).

C: Accuracy vs. Threshold is a threshold-sweep plot, not ROC.

Question

What does AUC primarily measure?

- A. Overall prediction accuracy
- B. Calibration quality
- C. Ability to distinguish positives from negatives
- D. Number of correct classifications

Answer: C

Ability to distinguish positives from negatives

Explanation

AUC (Area Under the ROC Curve) measures discrimination—the model's ability to *separate* positives from negatives.

Probabilistic interpretation: AUC is the probability that a randomly chosen positive example receives a higher predicted score than a randomly chosen negative example.

Why the others are wrong:

- A: AUC is not accuracy. A model can have high AUC but low accuracy at a specific threshold (and vice versa).
- B: Calibration measures whether predicted probabilities match observed frequencies—a different concept from discrimination.
- D: AUC is threshold-independent and does not count correct classifications directly.

Key point: AUC is about *ranking*, not about correctness at a single operating point.

Question

An AUC of 0.50 suggests:

- A. Perfect classification
- B. Random guessing
- C. Severe overfitting
- D. High precision

Answer: B

Random guessing

Explanation

An AUC of 0.50 means the ROC curve lies on the diagonal line ($TPR = FPR$). The model has no discrimination ability—it performs exactly like random guessing.

AUC scale:

0.50 = random guessing (no discrimination)

0.50–0.70 = poor

0.70–0.80 = acceptable

0.80–0.90 = excellent

0.90–1.00 = outstanding

1.00 = perfect separation

Why the others are wrong:

A: Perfect classification corresponds to $AUC = 1.00$.

C: $AUC = 0.50$ says nothing about overfitting; it means no signal.

D: AUC does not measure precision.

Question

Which AUC value corresponds to a perfect classifier?

- A. 0.50
- B. 0.75
- C. 0.90
- D. 1.00

Answer: D

1.00

Explanation

An AUC of 1.00 means the ROC curve passes through the top-left corner ($\text{TPR} = 1$, $\text{FPR} = 0$). The model perfectly separates positives from negatives at some threshold.

Why the others are wrong:

- A (0.50): Random guessing.
- B (0.75): Acceptable but not perfect.
- C (0.90): Excellent but not perfect.

Note: In practice, $\text{AUC} = 1.00$ usually indicates a data leak, label leakage, or an overly easy problem—real-world models rarely achieve it.

Question

Why is AUC often preferred to accuracy?

- A. It avoids the need for labels
- B. It evaluates performance across all thresholds
- C. It measures clustering quality
- D. It measures calibration

Answer: B

It evaluates performance across all thresholds

Explanation

Accuracy depends on a single threshold (usually 0.5). AUC summarizes performance across all possible thresholds, so it is threshold-independent.

Why this matters:

You can compare models without committing to a threshold.

AUC is robust to class imbalance (unlike accuracy).

It reflects the model's ranking quality overall.

Why the others are wrong:

A: AUC still requires ground-truth labels.

C: Clustering is unsupervised; AUC is for classification.

D: Calibration is a separate concept (predicted probability vs. observed frequency).

Question

Two models have AUC values of 0.91 and 0.73.
What can be concluded?

- A. The first model has better discrimination ability
- B. The first model has higher precision
- C. The first model has higher recall
- D. The first model has higher accuracy

Answer: A

The first model has better discrimination ability

Explanation

AUC measures discrimination / ranking ability. Since $0.91 > 0.73$, the first model separates positives from negatives better.
Why the others are wrong:

B: AUC says nothing about precision at a specific threshold.

C: AUC says nothing about recall at a specific threshold.

D: AUC is not accuracy; a model with higher AUC can have lower accuracy at a poorly chosen threshold.

Key point: AUC compares *overall ranking quality*, not any single threshold metric like precision, recall, or accuracy.

Question

An AUC of 0.85 means approximately:

- A. 85% of predictions are correct
- B. 85% recall
- C. An 85% chance a randomly chosen positive receives a higher score than a randomly chosen negative
- D. An 85% false positive rate

Answer: C

An 85% chance a randomly chosen positive receives a higher score than a randomly chosen negative

Explanation

This is the standard probabilistic interpretation of AUC:

$$AUC = P(\text{score of a random positive} > \text{score of a random negative})$$

An AUC of 0.85 means that in 85% of randomly drawn (positive, negative) pairs, the model assigns the positive a higher score.

Why the others are wrong:

- A: 85% accuracy is unrelated to $AUC = 0.85$.
- B: AUC is not recall.
- D: AUC is not FPR; FPR is a single-threshold quantity.

Memory aid: AUC = probability of correct ranking of a random pair.

Question

A highly imbalanced fraud dataset contains 99% non-fraud observations. Which metric is often more informative than accuracy?

- A. AUC
- B. Mean
- C. Variance
- D. R-squared

Answer: A

AUC

Explanation

With 99% non-fraud, a trivial model that predicts “non-fraud” for every observation achieves 99% accuracy while catching zero fraud cases. This is the classic class imbalance trap.

AUC is more informative because:

- It measures discrimination across all thresholds.

- It is insensitive to the class distribution.

- A random classifier still gives $AUC = 0.50$, regardless of imbalance.

Why the others are wrong:

- B: Mean is a descriptive statistic, not a classifier metric.

- C: Variance is not a classification metric.

- D: R-squared is for regression, not classification.

Other useful metrics for imbalanced data: precision, recall, F1, and the Precision-Recall curve.

Question

Moving along an ROC curve is typically caused by:

- A. Changing the classification threshold
- B. Retraining the model
- C. Increasing sample size
- D. Changing the target variable

Answer: A

Changing the classification threshold

Explanation

Each point on an ROC curve corresponds to a specific classification threshold. As you slide the threshold from 0 to 1:

Lower threshold → more positives predicted → higher TPR and higher FPR (move up-right).

Higher threshold → fewer positives predicted → lower TPR and lower FPR (move down-left).

So moving along the curve = changing the threshold on a *fixed* model.

Why the others are wrong:

B: Retraining produces a *different* ROC curve (the whole curve shifts), not movement along the same curve.

C: Increasing sample size does not move you along the curve.

D: Changing the target variable changes the problem entirely.

Question

Which statement regarding AUC is TRUE?

- A. AUC equals accuracy
- B. AUC requires a single threshold
- C. AUC measures discrimination across thresholds
- D. AUC measures precision

Answer: C

AUC measures discrimination across thresholds

Explanation

AUC is threshold-independent: it summarizes the model's discrimination ability across *all* thresholds by integrating the area under the ROC curve. Why the others are wrong:

- A: $AUC \neq$ accuracy. They measure different things; accuracy is threshold-dependent, AUC is not.
- B: AUC requires *no* single threshold—that is its main advantage.
- D: AUC is not precision. Precision is a single-threshold metric.

The GARP trap: The distractors mix up AUC with accuracy, precision, and threshold-dependent metrics. Remember: $AUC =$ discrimination across all thresholds.